



Application Descriptions

7

Energy Management

80

HVAC

2

Summary

This document specifies the Functional Blocks for the control of HVAC appliances and energy management.

This document is WGI Approved.

Document updates

Version	Date	Modifications
1.0	2024_07_19	new
1.1	2024_10_15	All Chapter, correction according first comments of WGI members
1.2	2024_12_13	FB DHW refers to Appl Des 7/11/3 Minor changes caused by Comments of WGI
1.3	2024_12_17	Typos corrected
1.4	2025_01_02	Load flexible available and Power_flexible_Available deleted
1.5	2025_02_04 2025_02_04	Names of Datapoints changed, Font change to Arial 10. FB Name added to Headlines. Figure in Chapter 2 added 1 Typos corrected
1.6	2025_03_27	PIDs added
1.7	2025_06_24	Filename in foot line corrected. FB Heating rod, picture corrected (inputs were not mirrored in properties)
01.01.01	2025.08.04	Creation of the WGI Approved version.

References

None.

Filename: 07_80_02 HVAC for EMS v01.01.01.docx
Version: 01.01.01
Status: WGI approved
Savedate: 2025.08.04
Number of pages: 82

Contents

1	Application: HVAC for EMS	4
1.1	Introduction	4
1.2	Functional Block: Heat Pump	4
1.2.1	Introduction	4
1.2.2	Function Block Diagram: Heat Pump	6
1.2.3	List of Datapoints Heat Pump	7
1.2.4	Inputs, Outputs, Parameters and Diagnostic Data of Heat Pump	10
1.3	Functional Block: Solar heat control	43
1.3.1	Introduction	43
1.3.2	Function Block Diagram: Solar heat control	43
1.3.3	List of Datapoints: Solar heat control	44
1.3.4	Inputs, Outputs, Parameters and Diagnostic Data of Solar Heat Control	45
1.4	Functional Block: Heating rod	52
1.4.1	Introduction	52
1.4.2	Function Block Diagram: Heating rod	53
1.4.3	List of Datapoints: Heating Rod	54
1.4.4	Inputs, Outputs, Parameters and Diagnostic Data of Heating Rod	56
1.5	Functional Block: Buffer Storage	67
1.5.1	Function Block Diagram: Buffer Storage	67
1.5.2	List of Datapoints: Buffer Storage	68
1.5.3	Inputs, Outputs, Parameters and Diagnostic Data of Buffer Storage	70
1.6	Functional Block: Domestic Hot Water	81

1 Application: HVAC for EMS

1.1 Introduction

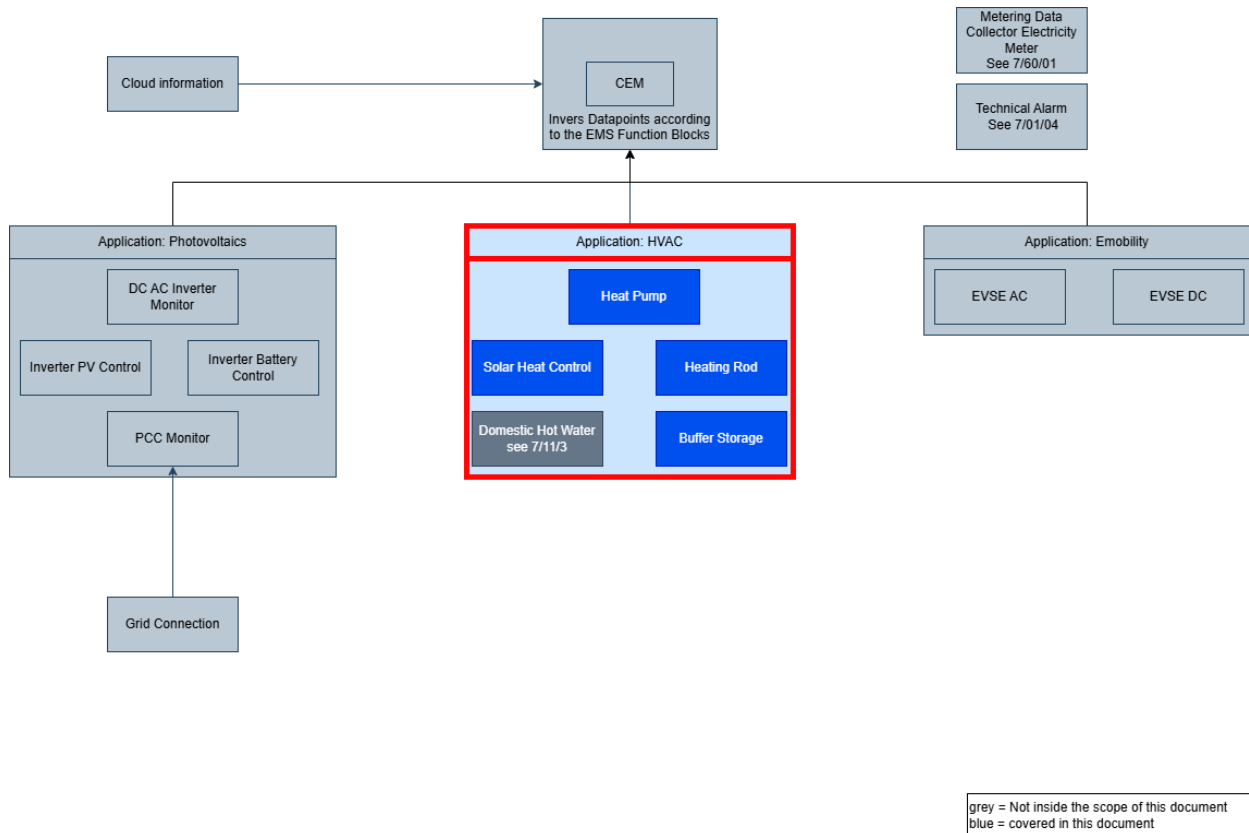


Figure 1 - Application "HVAC"

This Application consists of different functions

- Heat Pump to source energy for Heating, Cooling and Domestic Hot Water,
- Solar Heat Control to use local solar heated water,
- Heating Rod to support the heat pump at low outside temperature or in times when the local Solar PV has cheap energy,
- Buffer storage, where the energy is stored,
- Domestic Hot Water to store hot water for the people in the building.

1.2 Functional Block: Heat Pump

1.2.1 Introduction

A heat pump is designed to provide thermal energy from a reservoir at a lower temperature as useful heat at a higher temperature to a system to be heated.

A heat pump extracts thermal energy from the environment outside the building and uses it to heat the building space or to provide hot water.

Types of heat pumps are distinguished by the energy source e.g. air, soil, or groundwater.

In essence a heat pump makes use of the energy absorption and release by the phase transition of a refrigerant. The essential process steps are:

- Evaporation: The evaporator absorbs ambient heat and transfers it to the refrigerant, which is initially liquid, causing it to evaporate.

- Compression: A compressor compresses the refrigerant (now a gas), causing the temperature to rise.
- Condensing: The refrigerant releases the heat in the condenser and becomes liquid again.
- Expanding: The refrigerant is expanded and returns to the evaporator to repeat the cycle.

The indicator of the heat pumps performance is called COP (Coefficient Of Performance). It is calculated by the ratio of provided heat power and electrical energy used. The COP can typically be found in the manufacturer's instruction.

Heat pumps are available in two designs, the monobloc and the split heat pump. The monobloc heat pump is a compact design in which all refrigerant components (evaporator, compressor, condenser, expansion valve) are combined inside one housing, which is usually installed outdoors.

Split design heat pumps consist of an outdoor unit and at least one indoor unit, which are connected to each other by refrigerant pipes. The outdoor unit contains the evaporator with the air blower and usually also the compressor. In contrast, the indoor unit - or several units in multi-split systems - only contains the condenser. In addition, the control unit is usually connected to the indoor unit.

A heat pump system comprises various components that may be influenced by the EMS:

The heat pump itself as heat source, one or several buffers for storing the heating media and for storing hot water. Valves and pumps are controlling the water flow between the different components. A heating rod may be used as an auxiliary heating source. The control is mainly done via temperature setpoints of the system. For feedback purposes the temperatures at the relevant system components are measured and may be used in the control loop: Source temperature, water flow temperature, return flow temperature, buffer water temperature.

The heat pump system may have inherent latencies to operate effectively. This may lead to control operations that are executed with delay. The energy management system should consider these latencies and check details in the manufacturer's instructions. The heat pump FB provides diagnose objects to indicate the remaining run time.

This function block is representing the control of a heat pump and providing the objects for setpoint control and monitoring of the relevant sensor values.

1.2.2 Function Block Diagram: Heat Pump

Heat Pump			
Inputs	Parameters	Descriptions	
HeatPumpMode	HeatPumpMode		
TempFlowWaterSetp	TempFlowWaterSetp		
ElectricalPowerLimit	ElectricalPowerLimit		
TempOutside	TempOutside		
DHWMMode	DHWMMode		
TempSetpExtBuffEff	TempSetpExtBuffEff		
TempWaterExtBuff	TempWaterExtBuff		
		EnergyEfficiency	EnergyEfficiency
		HeatPumpModeEff	HeatPumpModeEff
		DHWMModeEff	DHWMModeEff
		ErrorHeatPump	ErrorHeatPump
		TempFlowWater	TempFlowWater
		TempFlowWaterSetpEff	TempFlowWaterSetpEff
		TempReturnWater	TempReturnWater
		TempSetpExtBuff	TempSetpExtBuff
		TempWaterBuff	TempWaterBuff
		TempSrc	TempSrc
		EnergyElectricalTotal	EnergyElectricalTotal
		EnergyElectricalHeat	EnergyElectricalHeat
		EnergyElectricalCool	EnergyElectricalCool
		EnergyElectricalDHW	EnergyElectricalDHW
		EnergyThermalTotal	EnergyThermalTotal
		EnergyThermalHeat	EnergyThermalHeat
		EnergyThermalCool	EnergyThermalCool
		EnergyThermalDHW	EnergyThermalDHW
		CurrentElectricalPowerTotal	CurrentElectricalPowerTotal
		CurrentThermalPowerTotal	CurrentThermalPowerTotal
		OpHrsCompressor	OpHrsCompressor
		CountCompressorStarts	CountCompressorStarts
	PowerLimitMaxGridSup		
	TempBufferBoost		
	TempBufferNormal		
		PowerLimitMax	
		LockTimeMin	
		RunTimeMin	

Figure 2 – Heatpump functional block

1.2.3 List of Datapoints Heat Pump

Inputs		
Datapoint	Description/Remarks	Datapoint Type
HeatPumpMode	With the operation mode heat pump Command the Energy management system controls the heat pump. The modes are described in clause 1.2.4.5. Modes 1.4 are used and well known for the "SG Ready" contacts of heat pumps. The description of the function of these modes can be found in the internet and are very common for Heat pumps.	20.1217 DPT_HeatpumpMode
TempFlowWaterSetp	Setpoint for the flow water temperature	9.001 DPT_Value_Temp
ElectricalPowerLimit	As an alternative to the control via the operation mode heat pump command, the heat pump could be controlled by the limitation of the electrical power. For that the "Operation Mode Heat pump Command " shall be "0".	14.056 DPT_Value_Power
TempOutside	Present outdoor temperature	9.001 DPT_Value_Temp
DHWMode	The operation mode heat pump DHW is used by the EMS to control the temperature setpoint of the domestic hot water. Modes are Comfort, Standby, Eco, Frost, Legionella protect	20.103 DPT_DHWMode
TempSetpExtBuffEff	This is the water temperature setpoint of an external buffer storage. This data point could be necessary to change the setpoint of this buffer storage for energy management reasons	9.001 DPT_Value_Temp
TempWaterExtBuff	This is the present water temperature of an external buffer storage. If there are temperatures of different layers available, typically the temperature of the top layer is used.	9.001 DPT_Value_Temp

Outputs		
Datapoint	Description/Remarks	Datapoint Type
EnergyEfficiency	This is the ratio between used electrical Energy and thermal Energy. This value is also known as COP Coefficient of Performance	9.031 DPT_Coefficient
HeatPumpModeEff	With the operation mode heat pump Command the Energy management system controls the heat pump. The modes are described clause 1.2.4.5. Modes 1.4 are used and well known for the "SG Ready" contacts of heat pumps. The description of the function of these modes can be found in the internet and are very common for Heat pumps.	20.1217 DPT_HeatpumpMode

Outputs		
Datapoint	Description/Remarks	Datapoint Type
DHWModeEff	The operation mode heat pump DHW status used to control the functional block Domestic Hot Water Modes are Comfort, Standby, Eco, Frost, Legionella protect	20.103 DPT_DHWMode
ErrorHeatPump	Global error message of the battery where true means "There is an error" and false means "There is no error". For detailed information about errors or alarms the function block "technical alarm" shall be used, see KNX Standard 07_01_04 Technical Alarm	1.005 DPT_Alarm
TempFlowWater	Present flow water temperature measured at the outlet of the heat pump	9.001 DPT_Value_Temp
TempFlowWaterSetpEff	State of the setpoint of the flow water temperature	9.001 DPT_Value_Temp
TempReturnWater	Return water temperature measured at the inlet of the heat pump	9.001 DPT_Value_Temp
TempSetpExtBuff	This is the water temperature setpoint for an external buffer storage. With this data point the heat pump can control an external buffer e.g. to increase the stored energy in the external buffer	9.001 DPT_Value_Temp
TempWaterBuff	Present temperature of the internal buffer storage of the heat pump, if available	9.001 DPT_Value_Temp
TempSrc	Present temperature of the source medium measured at the inlet of the heat pump	9.001 DPT_Value_Temp
EnergyElectricalTotal	Sum of the consumed energy for heat and cooling production including auxiliary units since installation of the heat pump	13.010 DPT_ActiveEnergy
EnergyElectricalHeat	Sum of the consumed energy for heat production including auxiliary units since installation of the heat pump	13.010 DPT_ActiveEnergy
EnergyElectricalCool	Sum of the consumed energy for cooling production including auxiliary units since installation of the heat pump	13.010 DPT_ActiveEnergy
EnergyElectricalDHW	Sum of the consumed energy including auxiliary units for heating of domestic hot water since installation of the heat pump	13.010 DPT_ActiveEnergy
EnergyThermalTotal	Sum of the produced thermal energy since installation of the heat pump	13.010 DPT_ActiveEnergy
EnergyThermalHeat	Sum of the produced thermal energy for heating since installation of the heat pump	13.010 DPT_ActiveEnergy
EnergyThermalCool	Sum of the produced thermal energy for cooling since installation of the heat pump	13.010 DPT_ActiveEnergy
EnergyThermalDHW	Sum of the produced thermal energy for heating of domestic hot water since installation of the heat pump	13.010 DPT_ActiveEnergy
CurrentElectricalPowerTotal	Present consumed electrical power of the compressor	14.056 DPT_Value_Power
CurrentThermalPowerTotal	present provided thermal power of the heat pump	14.056 DPT_Value_Power

Outputs		
Datapoint	Description/Remarks	Datapoint Type
OpHrsCompressor	Total operation time of the compressor since installation of the heat pump	13.100 DPT_LongDeltaTimeSec
CountCompressorStarts	Number of starts of the compressor since installation of the heat pump.	7.001 DPT_Value_2_Ucount

Parameters		
Datapoint	Description/Remarks	Datapoint Type
PowerLimitMaxGridSuppMode	Maximum of the power limit during Grid Supporting Mode (Mode 6)	14.056 DPT_Value_Power
TempBufferBoost	Maximum of the limit of buffer temperature during forced mode (Mode 3,4)	9.001 DPT_Value_Temp
TempBufferNormal	Setpoint for buffer temperature in normal mode (Mode 1)	9.001 DPT_Value_Temp

Diagnostic Data		
Datapoint	Description/Remarks	Datapoint Type
PowerLimitMax	Maximum of the limit of power of the heat pump	14.056 DPT_Value_Power
LockTimeMin	Value for the minimum lock time in seconds	7.005 DPT_TimePeriodSec
RunTimeMin	Value for the minimum run time in seconds	7.005 DPT_TimePeriodSec

1.2.4 Inputs, Outputs, Parameters and Diagnostic Data of Heat Pump

1.2.4.1 Inputs: Heat Pump

1.2.4.1.1 HeatPumpMode

DP Name:	HeatPumpMode	Abbr.:		Mandatory	☒
FB Name:	Heat pump			Can be internal	☐
Description					
With the operation mode heat pump command the Heat pump is controlled by the Energy management system. Modes 1..4 are used and well known for the "SG Ready" contacts of heat pumps. The description of the function of these modes can be found in the internet and are very common for Heat pumps. In Grid supporting Mode the power consumption of the heat pump is limited to the value Power, AC limited. In Mode "function deactivated" the Heat Pump is no more controlled by the datapoint operation mode heat pump Command but by other mechanisms e.g. Power, electrical, limitation					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_HeatpumpMode	DPT_ID:	20.1217		
Field		Supp.	Range	Unit	Default
Field1	Setting the heat pump mode 0 = DP deactivated 1 = blocking for max 2 h 2 = normal 3 = boost 4 = kick in 5: = Off 6 = Grid supporting mode 7 to 255 = not used; reserved	M	Full	none	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		None			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	51	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.1.2 TempFlowWaterSetp

DP Name:	TempFlowWaterSetp	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Setpoint for the flow water temperature					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Setting the setpoint for the flow water temperature	M	5-95°C	°C	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:	None				
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	52	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Value below or above the limit must be ignored					
Special Features					

1.2.4.1.3 ElectricalPowerLimit

DP Name:	ElectricalPowerLimit	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
As an alternative to the control via the operation mode heat pump command, the heat pump could be controlled by the limitation of the electrical power. For that the "operation mode heat pump command " shall be "0"					
Datapoint Type					
DPT Name:					
DPT Format:	DPT Value Power	DPT ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Setting the electrical power limit	M	1-20 kW	W	4,2 kW
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous Request	☒	Cyclically:	☐	Time-out:	
	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		None			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	53	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Value below or above the limit must be set to the limit					
Special Features					

1.2.4.1.4 TempOutside

DP Name:	TempOutside	Abbr.:		Mandatory	☒
FB Name:	Heat pump			Can be internal	☐
Description					
Present outdoor temperature					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Setting the present outdoor temperature	M	-50 to 60	°C	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	54	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Values below or above the limit shall be ignored					
Special Features					

1.2.4.1.5 DHWMode

DP Name:	DHWMode		Abbr.:		Mandatory	☐
FB Name:	Heat pump				Can be internal	☐
Description						
The operation mode heat pump DHW is used by the EMS to control the temperature setpoint of the domestic hot water. Modes are Comfort, Standby, Eco, Frost, Legionella protect						
Datapoint Type						
DPT Name:						
DPT Format:	DPT_DHWMode		DPT ID:	20.103		
Field		Supp.	Range	Unit	Default	
Field1	Setting the DHW operation mode of the heat pump 0 = Auto 1 = LegioProtect 2 = Normal 3 = Reduced 4 = Off/FrostProtect 5 ... 255 = reserved	M	Full	none	none	
Access Type						
◆ Input						
	N → this	☒	1 → this	☐		
	Spontaneous	☒	Cyclically:	☐	Time-out:	
	Request	☐	Polling:	☐	Period:	
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	none				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	139	Property ID:	55	
		Start-Index:	1	N° of elements	1	
	Property access:	Read only	☐	Read/Write	☒	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐	
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

1.2.4.1.6 TempSetpExtBuffEff

DP Name:	TempSetpExtBuffEff	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Temperature setpoint of an external buffer storage. This data point could be necessary to change the setpoint of this buffer storage for energy management reasons					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Setting the temperature setpoint of an external buffer storage	M	5 to 95	°C	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	none			
◆ Property					Mandatory: ☒
	DP Address: (in the server)	IO Type(ID):	139	Property ID:	56
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
		Saved value:	☐	Current value (not for input):	
		Transmit on bus (only for output):	☐	Read from bus (only for input):	
Exception Handling					
Value below or above are set to the limit					
Special Features					

1.2.4.1.7 TempWaterExtBuff

DP Name:	TempWaterExtBuff	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☒
Description					
Present temperature of an external buffer storage. If there are temperatures of different layers available, typically the temperature of the top layer is used.					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Setting the present temperature of an external buffer storage	M	5 to 95	°C	none
Access Type					
◆ Input					
	N → this	☒	1 → this	☐	
	Spontaneous	☒	Cyclically:	☐	Time-out:
	Request	☐	Polling:	☐	Period:
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	none			
◆ Property					Mandatory: ☒
	DP Address: (in the server)	IO Type(ID):	139	Property ID:	57
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☐	Read/Write	☒
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
		Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐
Exception Handling					
Value below or above the limit must be ignored					
Special Features					

1.2.4.2 Outputs: Heat Pump

1.2.4.2.1 EnergyEfficiency

DP Name:	EnergyEfficiency	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
The ratio between used electrical energy and thermal energy. This value is also known as COP Coefficient of Performance					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Coefficient	DPT_ID:	9.031		
Field		Supp.	Range	Unit	Default
Float	Value representing the ratio between used electrical and thermal energy	M	full	none	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		none			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	60	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.2 HeatPumpModeEff

DP Name:	HeatPumpModeEff	Abbr.:		Mandatory	☒
FB Name:	Heat pump			Can be internal	☐
Description					
The operation mode heat pump state shows the state of the heat pump. Modes 1.4 are used and well known for the ""SG Ready"" contacts of heat pumps. The description of the function of these modes can be found on the internet and are very common for Heat pumps. In grid supporting mode the power consumption of the heat pump is limited to the value Power, AC limited.					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_HeatpumpMode	DPT_ID:	20.1217		
Field		Supp.	Range	Unit	Default
Field1	Value reflecting the current heat pump operation mode	M	full	none	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	61	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.3 DHWModeEff

DP Name:	DHWModeEff		Abbr.:		Mandatory	☐																						
FB Name:	Heat pump				Can be internal	☐																						
Description																												
The operation mode heat pump DHW status is used to control the function block Domestic Hot Water Modes are Comfort, Standby, Eco, Frost, Legionella protection																												
Datapoint Type																												
DPT Name:																												
DPT Format:	DPT_DHWMode		DPT ID:	20.103																								
Field		Supp.	Range	Unit	Default																							
Field1	Value informing about the DHW operation mode of the heat pump 0 = Auto 1 = LegioProtect 2 = Normal 3 = Reduced 4 = Off/FrostProtect 5 ... 255 = reserved	M	full	none	none																							
Access Type																												
◆ Output																												
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☒</td> </tr> <tr> <td></td> <td></td> <td>Δ-Value:</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Period:</td> <td></td> </tr> <tr> <td>Request</td> <td>☒</td> <td></td> <td></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☒			Δ-Value:				Cyclic	☐			Period:		Request	☒			Min repetition time:			
this → M	☒	this → 1	☐																									
Spontaneous	☒	COV:	☒																									
		Δ-Value:																										
		Cyclic	☐																									
		Period:																										
Request	☒																											
Communication Type																												
◆ Group Object Datapoint					Mandatory:	☒																						
Default Group Address:		none																										
◆ Property					Mandatory:	☒																						
DP Address: (in the server)	IO Type(ID):	139	Property ID:	62																								
	Start-Index:	1	N° of elements	1																								
Property access:	Read only	☒	Read/Write	☐																								
Protection *)	Read level	-	Write level	-																								
Dynamics																												
Power down:	Save:	☐																										
Power up:	Value:	No initialisation:	☒	Default value:	☐																							
		Saved value:	☐	Current value (not for input):	☐																							
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																							
Exception Handling																												
Special Features																												

1.2.4.2.4 ErrorHeatPump

DP Name:	ErrorHeatPump	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Global error message of the battery where "true" means "There is an error" and "false" means "There is no error". For detailed error or alarm messages the functional block "technical alarm" shall be used, see KNX Standard 07_01_04.					
Datapoint Type					
DPT Name:					
DPT Format:	DPT Alarm	DPT ID:	1.005		
Field		Supp.	Range	Unit	Default
B	Value informing about the error status of the heat pump (0 = no error, 1 = error)	M	full	bool	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		none			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	63	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.5 TempFlowWater

DP Name:	TempFlowWater	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Present flow water temperature measured at the outlet of the heat pump					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present flow water temperature at the heat pump outlet	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	64	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.6 TempFlowWaterSetpEff

DP Name:	TempFlowWaterSetpEff	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Feedback of the setpoint for the flow water temperature					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value informing about the setpoint for the flow water temperature	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	65	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

1.2.4.2.7 TempReturnWater

DP Name:	TempReturnWater	Abbr.:		Mandatory	☒
FB Name:	Heat pump			Can be internal	☐
Description					
Return water temperature measured at the inlet of the heat pump					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value informing about the return water temperature measured at the heat pump inlet	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	66	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.8 TempSetpExtBuff

DP Name:	TempSetpExtBuff		Abbr.:		Mandatory	☐
FB Name:					Can be internal	☐
Description						
Temperature setpoint for an external buffer storage. With this data point the heat pump can control an external Buffer e.g. to increase the stored energy in the external buffer						
Datapoint Type						
DPT Name:			DPT ID:	9.001		
DPT Format:	DPT_Value_Temp		Supp.	Range	Unit	Default
Field			M	full	°C	none
Float	Value informing about the temperature setpoint for an external buffer storage					
Access Type						
◆ Output						
this → M	☒	this → 1	☐			
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %	Min repetition time:
		Cyclic	☐	Period:		
Request	☒					
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:		None				
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	67		
	Start-Index:	1	N° of elements	1		
Property access:	Read only	☒	Read/Write	☐		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

1.2.4.2.9 TempWaterBuff

DP Name:	TempWaterBuff	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Present temperature of the internal buffer of the heat pump, if available					
Datapoint Type					
DPT_Name:					
DPT_Format:	DPT_Value_Temp	DPT_ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present temperature of the internal heat pump buffer	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	68	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

1.2.4.2.10 TempSrc

DP Name:	TempSrc	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Present temperature of the source medium measured at the inlet of the heat pump					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present temperature of the source medium measured at the heat pump inlet	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		None			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	69	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.11 EnergyElectricalTotal

DP Name:	EnergyElectricalTotal		Abbr.:		Mandatory	☐
FB Name:	Heat pump				Can be internal	☐
Description						
Sum of the consumed energy for heat and cooling production including auxiliary units since installation of the heat pump						
Datapoint Type						
DPT_Name:						
DPT Format:	DPT_ActiveEnergy		DPT_ID:	13.010		
Field			Supp.	Range	Unit	Default
Signed Value	Value informing about the sum of consumed energy for heat and cooling production since installation		M	full	Wh	none
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	None				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	139	Property ID:	70	
		Start-Index:	1	N° of elements	1	
	Property access:	Read only	☒	Read/Write	☐	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
			Saved value:	☐	Current value (not for input):	☐
		Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling						
Special Features						

1.2.4.2.12 EnergyElectricalHeat

DP Name:	EnergyElectricalHeat	Abbr.:		Mandatory	☐
FB Name:				Can be internal	☐
Description					
Sum of the consumed energy for heat production including auxiliary units since installation of the heat pump					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_ActiveEnergy	DPT_ID:	13.010		
Field		Supp.	Range	Unit	Default
Signed Value	Value informing about the sum of the consumed energy for heat production since installation	M	full	Wh	none
Access Type					
◆ Output					
	this → M	☒	this → 1	☐	
	Spontaneous	☒	COV:	☒	Δ-Value: >1 %
			Cyclic	☐	Period:
	Request	☒			
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	None			
◆ Property					Mandatory: ☒
	DP Address: (in the server)	IO Type(ID):	139	Property ID:	71
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☒	Read/Write	☐
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):		☐	Read from bus (only for input):
Exception Handling					
Special Features					

1.2.4.2.13 EnergyElectricalCool

DP Name:	EnergyElectricalCool	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Sum of the consumed energy for cooling production including auxiliary units since installation of the heat pump					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_ActiveEnergy	DPT_ID:	13.010		
Field		Supp.	Range	Unit	Default
Signed Value	Value informing about the sum of the consumed energy for cooling production since installation	M	full	Wh	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	72	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.14 EnergyElectricalDHW

DP Name:	EnergyElectricalDHW	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Sum of the consumed energy including auxiliary units for heating of domestic hot water since installation of the heat pump					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_ActiveEnergy	DPT_ID:	13.010		
Field		Supp.	Range	Unit	Default
Signed Value	Value informing about the sum of the consumed energy for heating of domestic hot water since installation	M	full	Wh	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	73	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.15 EnergyThermalTotal

DP Name:	EnergyThermalTotal	Abbr.:		Mandatory	☐
FB Name:				Can be internal	☐
Description					
Sum of the produced thermal energy since installation of the heat pump					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_ActiveEnergy	DPT ID:	13.010		
Field		Supp.	Range	Unit	Default
Signed Value	Value informing about the sum of the produced thermal energy since installation	M	full	Wh	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	74	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.16 EnergyThermalHeat

DP Name:	EnergyThermalHeat	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Sum of the produced thermal energy for heating since installation of the heat pump					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_ActiveEnergy	DPT ID:	13.010		
Field		Supp.	Range	Unit	Default
Signed Value	Value informing about the sum of the produced thermal energy for heating since installation	M	full	Wh	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	75	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.17 EnergyThermalCool

DP Name:	EnergyThermalCool	Abbr.:		Mandatory	☐																								
FB Name:	Heat pump			Can be internal	☐																								
Description																													
Sum of the produced thermal energy for cooling since installation of the heat pump																													
Datapoint Type																													
DPT_Name:																													
DPT_Format:	DPT_ActiveEnergy	DPT_ID:	13.010																										
Field		Supp.	Range	Unit	Default																								
Signed Value	Value informing about the sum of the produced thermal energy for cooling since installation	M	full	Wh	none																								
Access Type																													
◆ Output																													
<table border="1"> <tr> <td>this → M</td> <td>☒</td> <td>this → 1</td> <td>☐</td> </tr> <tr> <td>Spontaneous</td> <td>☒</td> <td>COV:</td> <td>☒</td> </tr> <tr> <td></td> <td></td> <td>Δ-Value:</td> <td>>1 %</td> </tr> <tr> <td></td> <td></td> <td>Cyclic</td> <td>☐</td> </tr> <tr> <td></td> <td></td> <td>Period:</td> <td></td> </tr> <tr> <td>Request</td> <td>☒</td> <td></td> <td></td> </tr> </table>	this → M	☒	this → 1	☐	Spontaneous	☒	COV:	☒			Δ-Value:	>1 %			Cyclic	☐			Period:		Request	☒							
this → M	☒	this → 1	☐																										
Spontaneous	☒	COV:	☒																										
		Δ-Value:	>1 %																										
		Cyclic	☐																										
		Period:																											
Request	☒																												
Communication Type																													
◆ Group Object Datapoint					Mandatory: ☒																								
Default Group Address:		None																											
◆ Property					Mandatory: ☒																								
DP Address: (in the server)	IO Type(ID):	139	Property ID:	76																									
	Start-Index:	1	N° of elements	1																									
Property access:	Read only	☒	Read/Write	☐																									
Protection *)	Read level	-	Write level	-																									
Dynamics																													
Power down:	Save:	☐																											
Power up:	Value:	No initialisation:	☒	Default value:	☐																								
		Saved value:	☐	Current value (not for input):	☐																								
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐																								
Exception Handling																													
Special Features																													

1.2.4.2.18 EnergyThermalDHW

DP Name:	EnergyThermalDHW	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Sum of the produced thermal energy for heating of domestic hot water since installation of the heat pump					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_ActiveEnergy	DPT_ID:	13.010		
Field		Supp.	Range	Unit	Default
Signed Value	Value informing about the sum of the produced thermal energy for heating of domestic hot water since installation	M	full	Wh	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	77	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

1.2.4.2.19 CurrentElectricalPowerTotal

DP Name:	CurrentElectricalPowerTotal	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Present consumed electrical power of the compressor					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Power	DPT ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present consumed electrical power of the compressor	M	full	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	78	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.20 CurrentThermalPowerTotal

DP Name:	CurrentThermalPowerTotal	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Present provided thermal power of the heat pump					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Power	DPT ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present provided thermal power	M	full	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	79	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.21 OpHrsCompressor

DP Name:	OpHrsCompressor	Abbr.:		Mandatory	☐
FB Name:				Can be internal	☐
Description					
Operation time of the compressor since installation of the heat pump					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_LongDeltaTimeSec	DPT_ID:	13.100		
Field		Supp.	Range	Unit	Default
Signed Value	Value informing about the operation time of the compressor since installation	M	full	s	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	78	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.2.22 CountCompressorStarts

DP Name:	CountCompressorStarts	Abbr.:		Mandatory	☐
FB Name:	Heat pump			Can be internal	☐
Description					
Number of starts of the compressor since installation of the heat pump.					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_2_Ucount	DPT ID:	7.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Value informing about the number of starts of the compressor since installation	M	full	none	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		None			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	139	Property ID:	79	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.2.4.3 Parameters: Heat Pump

1.2.4.3.1 PowerLimitMaxGridSuppMode

FB:	Heat pump	Property Name (Server):	PowerLimitMaxGridSuppMode			Mandatory	☐	Optional	☒
Description:									
Maximum of the limit of power during Grid Supporting Mode (Mode 6)									
DPT:	Name	DPT_Value_Power	DPT_ID	14.056	Datatype format:				
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Float	Parameter setting the maximum of the power limit during mode 6		M	1 to 20 kW	W	--	4,2 kW		
Communication:									
DP Address: (in the server)		Object Type:	139	Property ID:		111			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☐	-	Read/Write ☒					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐									
Value below or above the limit shall be set to the limit									
Special Features:									
None.									

1.2.4.3.2 TempBuffer Boost

FB:	Heat pump	Property Name (Server):	TempBufferBoost			Mandatory	☐	Optional	☒
Description:									
Maximum of the limit of Buffer Temperature during forced mode (Mode 3,4)									
DPT:	Name	DPT_Value_Temp	DPT_ID	9.001	Datatype format:				
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Float	Parameter setting the maximum of the limit of the buffer temperature during mode 3, 4		M	5 to 95	°C	0,01	none		
Communication:									
DP Address: (in the server)		Object Type:	139	Property ID:		112			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☐	-	Read/Write ☒					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐									
Value below or above the limit shall be set to the limit									
Special Features:									
None.									

1.2.4.3.3 TempBufferNormal

FB:	Heat pump	Property Name (Server):	TempBufferNormal			Mandatory	☐	Optional	☒
Description:									
Setpoint for buffer temperature in normal mode (Mode 1)									
DPT:	Name	DPT_Value_Temp	DPT_ID	9.001	Datatype format:				
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Float	Parameter setting the setpoint for the temperature in mode 1		M	5 to 95	°C	0,01	none		
Communication:									
DP Address: (in the server)		Object Type:	139	Property ID:		113			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☐	-	Read/Write ☒					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐									
Value below or above the limit shall be set to the limit									
Special Features:									
None.									

1.2.4.4 Diagnostic Data: Heat Pump

1.2.4.4.1 PowerLimitMax

FB:	Heat pump	Property Name (Server):	PowerLimitMax			Mandatory	☐	Optional	☒
Description:									
Maximum of power of the heat pump as written in the data sheet									
DPT:	Name	DPT_Value_Power	DPT_ID	14.056	Datatype format:				
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Float	Value informing about the maximum power of the heat pump		M	Full	W	--	none		
Communication:									
DP Address: (in the server)		Object Type:	139	Property ID:		120			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.2.4.4.2 LockTimeMin

FB:	Heat pump	Property Name (Server):	LockTimeMin			Mandatory	☐	Optional	☒
Description:									
Value for the minimum lock time in seconds									
DPT:	Name	DPT_TimePeriodSec	DPT_ID	7.005	Datatype format:				
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Time Period	Value informing about the minimum lock time		M	Full	s	1	none		
Communication:									
DP Address: (in the server)		Object Type:	139	Property ID:		121			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.2.4.4.3 RunTimeMin

FB:	Heat pump	Property Name (Server):	RunTimeMin			Mandatory	☐	Optional	☒
Description:									
Value for the minimum run time in seconds									
DPT:	Name	DPT_TimePeriodSec	DPT_ID	7.005	Datatype format:				
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Time Period	Value informing about the minimum run time		M	Full	s	1	none		
Communication:									
DP Address: (in the server)		Object Type:	139	Property ID:		122			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.2.4.5 New Data Point Types

1.2.4.5.1 Heat Pump Mode

<u>Format:</u>	1 octet: N ₈								
octet nr.	1								
field names	<table border="1"><tr><td><i>field1</i></td></tr></table>	<i>field1</i>							
<i>field1</i>									
encoding	<table border="1"><tr><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td><td>N</td></tr></table>	N	N	N	N	N	N	N	N
N	N	N	N	N	N	N	N		
<u>Encoding:</u>	Encoding absolute value N = [0 ... 255]								
<u>Unit:</u>	None								
<u>Resol.:</u>	None								
<u>PDT:</u>	PDT_ENUM8 (alt: PDT_UNSIGNED_CHAR)								

Datapoint Types				
<u>ID:</u>	<u>Name:</u>	<u>Encoding:</u>	<u>Range:</u>	<u>Use:</u>
20.1217	DPT_HeatpumpMode	<i>field1</i> = Heat pump Mode 0 = DP deactivated 1 = blocking for max 2 h 2 = normal 3 = boost 4 = kick in 5 = Off 6 = Grid supporting mode 7 to 255 = not used; reserved	[0 to 6]	FB

Modes 1..4 are used and well known for the "SG Ready" contacts of heat pumps. The description of the function of these modes can be found in the internet and they are very common for Heat pumps. In Grid supporting Mode the power consumption of the heat pump is limited to the value Power, AC limited.

This datapoint can be used for SGReady Actuators, Modes 1-4 are compatible to SGReady contacts.

1.3 Functional Block: Solar heat control

1.3.1 Introduction

A thermal solar system converts light into heat and then transfers this heat via pipes for use for the heating system or domestic hot water system. A dedicated circuit using frost protected fluid is transporting the collected energy controlled by a pump or valves via an exchanger into the domestic hot water system.

This functional block represents the control logic of a solar heat controller e.g. temperature sensor and pump or valve control.

1.3.2 Function Block Diagram: Solar heat control

[illegible]

Figure 3 – Solar Heat Control functional block

1.3.3 List of Datapoints: Solar heat control

Inputs		
Datapoint	Description/Remarks	Datapoint Type
PowerOnOff	On/Off signal to switch the thermal Power supply of the Solar panel e.g. by opening or closing a valve or switching a pump on or off. On means no limitation of the thermal power (valve open or pump on), Off means no thermal power (valve closed off pump off).	1.001 DPT_Bool
PowerControlSetp	Continuous control of the power of a solar panel. Value 0 switches the power "off". Any other value switches the power on with the manufacturer defined setting of the heating value in the scale of 0 -100%.	5.001 DPT_Scaling

Outputs		
Datapoint	Description/Remarks	Datapoint Type
PowerControlEff	Percent value that indicates the provided thermal power as the speed of the pump or valve position.	5.001 DPT_Scaling
TempSolarCollector	Present Solar Collector temperature measured at the highest point of the solar panel	9.001 DPT_Value_Temp
ThermalPowerHeat	Present thermal power of the solar panel provided into the connected system.	14.056 DPT_Value_Power
ThermalEnergyHeat	Present thermal energy of the solar panel	13.010 DPT_ActiveEnergy

Diagnostic Data		
Datapoint	Description/Remarks	Datapoint Type
EnergyEfficiency	Energy efficiency of the solar panels (produced Peak Power/ radiated power) given by the datasheet of the solar panel	5.001 DPT_Scaling
ThermalEnergyHeat Max	Maximum of the thermal energy of the solar panel	13.010 DPT_ActiveEnergy
ThermalPowerHeat Max	Maximum of the thermal power of the solar panel	14.056 DPT_Value_Power

1.3.4 Inputs, Outputs, Parameters and Diagnostic Data of Solar Heat Control

1.3.4.1 Inputs: Solar heat control

1.3.4.1.1 PowerOnOff

DP Name:	PowerOnOff	Abbr.:		Mandatory	☒
FB Name:	Solar heat control			Can be internal	☐
Description					
On/Off signal to switch the thermal power supply of the solar panel e.g. by opening or closing a valve or switching a pump on or off. "On" means no limitation of the thermal power (valve open or pump on), "Off" means no thermal power (valve closed off pump off).					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Bool	DPT_ID:	1.001		
Field		Supp.	Range	Unit	Default
B	Value switching the thermal power supply 1 = no limitation of the thermal power 0 = no thermal power	M	full	Bool	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	140	Property ID:	51	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.3.4.1.2 PowerControlSetp

DP Name:	PowerControlSetp	Abbr.:		Mandatory	☐
FB Name:	Solar heat control			Can be internal	☐
Description					
Percent value to limit the thermal power supply of the solar panel. The thermal power is typically reduced by reducing the speed of the pump or the opening of a valve					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Scaling	DPT ID:	5.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Setting the limit of the thermal power supply	M	full	%	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous Request	☒	Cyclically:	☐	Time-out:	
	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	140	Property ID:	52	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.3.4.2 Outputs: Solar heat control

1.3.4.2.1 PowerControlEff

DP Name:	PowerControlEff	Abbr.:		Mandatory	☒
FB Name:	Solar heat control			Can be internal	☐
Description					
Percent value that indicates the provided thermal power as the speed of the pump or valve position. If there is no continuous control of the power, only the values 0% and 100 % are possible					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Scaling	DPT_ID:	5.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Value informing about the current provided thermal power	M	full	%	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		none			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	140	Property ID:	60	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.3.4.2.2 TempSolarCollector

DP Name:	TempSolarCollector		Abbr.:		Mandatory	☒
FB Name:	Solar heat control				Can be internal	☐
Description						
Present Solar Collector temperature measured at the highest point of the solar panel						
Datapoint Type						
DPT Name:						
DPT Format:	DPT_Value_Temp		DPT ID:	9.001		
Field			Supp.	Range	Unit	Default
Float	Value informing about the present solar collector temperature		M	full	°C	none
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	none				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	140	Property ID:	60	
		Start-Index:	1	N° of elements	1	
	Property access:	Read only	☒	Read/Write	☐	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐	
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

1.3.4.2.3 ThermalPowerHeat

DP Name:	ThermalPowerHeat		Abbr.:		Mandatory	☐
FB Name:	Solar heat control				Can be internal	☐
Description						
Present thermal power of the solar panel provided into the connected system.						
Datapoint Type						
DPT Name:						
DPT Format:	DPT_Value_Power		DPT ID:	14.056		
Field			Supp.	Range	Unit	Default
Float	Value informing about the present thermal power of the solar panel		M	full	W	none
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	none				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	140	Property ID:	62	
		Start-Index:	1	N° of elements	1	
	Property access:	Read only	☒	Read/Write	☐	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐	
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

1.3.4.2.4 ThermalEnergyHeat

DP Name:	ThermalEnergyHeat		Abbr.:		Mandatory	☐
FB Name:	Solar heat control				Can be internal	☐
Description						
Present thermal energy of the solar panel						
Datapoint Type						
DPT Name:			DPT ID:	13.010		
DPT Format:	DPT_ActiveEnergy					
Field			Supp.	Range	Unit	Default
Signed Value	Value informing about the present thermal energy of the solar panel		M	full	Wh	none
Access Type						
◆ Output						
this → M	☒	this → 1	☐			
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %	Min repetition time:
		Cyclic	☐	Period:		
Request	☒					
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
Default Group Address:	none					
◆ Property					Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	140	Property ID:	63		
	Start-Index:	1	N° of elements	1		
Property access:	Read only	☒	Read/Write	☐		
Protection *)	Read level	-	Write level	-		
Dynamics						
Power down:	Save:	☐				
Power up:	Value:	No initialisation:	☒	Default value:	☐	
		Saved value:	☐	Current value (not for input):	☐	
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

1.3.4.3 Parameter: Solar heat control

This function block has no parameters.

1.3.4.4 Diagnostic Data: Solar heat control

1.3.4.4.1 EnergyEfficiency

FB:		Property Name (Server):	EnergyEfficiency			Mandatory	☐	Optional	☒
Description:									
Energy efficiency of the solar panels according EN 50530 as provided by the manufacturer									
DPT:	Name	DPT_Scaling	DPT_ID	5.001	Datatype format:				
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Unsigned Value	Value informing about the energy efficiency of the solar panels		M	full	%	0,04	none		
Communication:									
DP Address: (in the server)		Object Type:	140	Property ID:		111			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.3.4.4.2 ThermalEnergyHeatMax

FB:		Property Name (Server):	ThermalEnergyHeatMax			Mandatory	☐	Optional	☒
Description:									
Maximum of the thermal energy of the solar panel									
DPT:	Name	DPT_ActiveEnergy	DPT_ID		Datatype format:		13.010		
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Signed Value	Value informing about the maximum of the thermal energy of the solar panels		M	full	Wh	1	none		
Communication:									
DP Address: (in the server)		Object Type:	140	Property ID:		112			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.3.4.4.3 ThermalPowerHeatMax

FB:		Property Name (Server):		Mandatory	☐	Optional	☒
Description:							
Maximum of the thermal power of the solar panel							
DPT:	Name	DPT Value Power	DPT_ID		Datatype format:	14.056	
Field	Description		Sup.	Range	Unit	Resol.:	Default
Float	Value informing about the maximum of the thermal power of the solar panel		M	full	W	--	none
Communication:							
DP Address: (in the server)		Object Type:	140	Property ID:		123	
		Start-Index:	1	N° of elements		1	
Property access:		Read only ☒	-	Read/Write ☐			
Protection		Read level	-	Write level		-	
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Special Features:							
None.							

1.4 Functional Block: Heating rod

1.4.1 Introduction

The heating rod is used as accessory to support a renewable energy source (when primary source) for time periods where heating energy supply is not efficiently possible via the primary source or when a surplus of electrical energy is available.

Heating rods are typically used in conjunction with solar thermal systems or heat pumps. As solar thermal system and heat pump-based systems may not operate efficiently under certain conditions, a heating rod can provide the required energy to fill gaps.

As heating systems generally feed a water-based buffer, the heating rod is directly connected or integrated into the buffer.

The heating rod uses electrical energy to heat the water in the buffer to the desired temperature. It consists of an insulated electrical conductor that converts electricity into heat energy. The heating element can be screwed or plugged into the water storage.

There are various available systems to control the heating rod. The simplest variant is an on/off based control. In this case the targeted temperature is gained by the time the heater is on, another variant can switch the power of the heating element up or down in stages. The most convenient variant has continuous control. This allows the output of the heating elements in the buffer to be flexibly adjusted to the current heat demand.

The electricity required is usually obtained from the grid operator or may be directly produced by the domestic photovoltaic system.

1.4.2 Function Block Diagram: Heating rod

[illegible]

Figure 4 – Solar Heat Control functional block

1.4.3 List of Datapoints: Heating Rod

Inputs		
Datapoint	Description/Remarks	Datapoint Type
LoadOnOff	The heating rod is controlled via switching commands On and Off. Value On(1) activates the load. Value Off(0) switches the load off.	1.001 DPT_Bool
LoadControlSetp	Continuous control of the heating rod. Value 0 switches off the load. Any other value switches on the load with the manufacturer defined setting of the heating value in the scale of 0-100%.	5.001 DPT_Scaling
BufferFull	This signal is sent from the buffer, where the heating rod is located. As long the buffer is full (1) the heating rod is switched off to avoid over temperature in the buffer.	1.001 DPT_Bool

Outputs		
Datapoint	Description/Remarks	Datapoint Type
LoadControlEff	Percent value indicates the operation status of the heating rod.	5.001 DPT_Scaling
PowerL1	The present power value for consumption (positive value) of the device at phase L1.	14.056 DPT_Value_Power
PowerL2	The present power value for consumption (positive value) of the device at phase L2.	14.056 DPT_Value_Power
PowerL3	The present power value for consumption (positive value) of the device at phase L3.	14.056 DPT_Value_Power
PowerL1L2L3	The present power value for consumption (positive value) of the device cumulated at all phase L1, L2, L3.	14.056 DPT_Value_Power

Parameters		
Datapoint	Description/Remarks	Datapoint Type
LoadMin	Low boundary of the load	14.056 DPT_Value_Power

Diagnostic Data		
Datapoint	Description/Remarks	Datapoint Type
PowerL1Max	The present maximum power value for consumption (positive value) of the device at phase L1.	14.056 DPT_Value_Power
PowerL2Max	The present maximum power value for consumption (positive value) of the device at phase L2.	14.056 DPT_Value_Power
PowerL3Max	The present maximum power value for consumption (positive value) of the device at phase L3.	14.056 DPT_Value_Power
PowerL1L2L3Max	The present maximum power value for consumption (positive value) of the device cumulated at all phase L1, L2, L3.	14.056 DPT_Value_Power
PowerL1L2L3Max	Energy efficiency i.e. heating load output / electrical load input	5.001 DPT_Scaling

1.4.4 Inputs, Outputs, Parameters and Diagnostic Data of Heating Rod

1.4.4.1 Inputs: Heating Rod

1.4.4.1.1 LoadOnOff

DP Name:	LoadOnOff	Abbr.:		Mandatory	☒
FB Name:	Heating Rod			Can be internal	☐
Description					
The heating rod is controlled via switching commands On and Off. Value On(1) activates the load. Value Off(0) switches the load off.					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Bool	DPT ID:	1.001		
Field		Supp.	Range	Unit	Default
b	Controlling the heating rod 1 = activation 0 = deactivation	M	full	Bool	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	141	Property ID:	51	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.4.4.1.2 LoadControlSetp

DP Name:	LoadControlSetp	Abbr.:		Mandatory	☐
FB Name:	Heating Rod			Can be internal	☐
Description					
Continuous control of the heating rod. Value 0 switches the load "off". Any other value switches on the load with the manufacturer defined setting of the heating value in the scale of 0-100%.					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Scaling	DPT ID:	5.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Continuous control of the heating rod	M	full	%	none
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous Request	☒	Cyclically:	☐	Time-out:	
	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	141	Property ID:	52	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.4.4.1.3

1.4.4.1.4 BufferFull

DP Name:	BufferFull	Abbr.:		Mandatory	☒
FB Name:	Heating Rod			Can be internal	☐
Description					
This datapoint indicates when TRUE that the buffer temperature has reached the value of the temperature setpoint of the water. To have a stable output this value remains TRUE as long as the buffer temperature is higher than the temperature setpoint of the water minus hysteresis of the buffer. If the buffer temperature is below the temperature setpoint of the water minus hysteresis of the buffer the value becomes FALSE and remains FALSE until the buffer temperature has reached the value of temperature setpoint of the water.					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Bool	DPT ID:	1.002		
Field		Supp.	Range	Unit	Default
b	Informing that the buffer temperature has reached the value of the temperature setpoint of the water (1) or not (0)	M	full	bool	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	141	Property ID:	53	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.4.4.2 Outputs: Heating Rod

1.4.4.2.1 LoadControlEff

DP Name:	LoadControlEff	Abbr.:		Mandatory	☐
FB Name:	Heating Rod			Can be internal	☐
Description					
Percent value indicating the operation status of the heating rod. If there is no continuous Control of the heating rod, only the values 0% and 100% are possible					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Scaling	DPT_ID:	5.001		
Field		Supp.	Range	Unit	Default
Unsigned Value	Value informing about the operation status of the heating rod	M	full	%	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		none			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	141	Property ID:	60	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.4.4.2.2 PowerL1

DP Name:	PowerL1	Abbr.:		Mandatory	☐
FB Name:	Heating Rod			Can be internal	☐
Description					
The present power value for consumption (positive value) of the device at phase L1.					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Power	DPT ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present power value for consumption at phase L1	M	full	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	141	Property ID:	61	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

1.4.4.2.3 PowerL2

DP Name:	PowerL2	Abbr.:		Mandatory	☐
FB Name:	Heating Rod			Can be internal	☐
Description					
The present power value for consumption (positive value) of the device at phase L2.					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Power	DPT ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present power value for consumption at phase L2	M	full	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	141	Property ID:	62	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

1.4.4.2.4 PowerL3

DP Name:	PowerL3	Abbr.:		Mandatory	☐
FB Name:	Heating Rod			Can be internal	☐
Description					
The present power value for consumption (positive value) of the device at phase L3.					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Power	DPT ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present power value for consumption at phase L3	M	full	W	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	141	Property ID:	63	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

1.4.4.2.5 PowerL1L2L3

DP Name:	PowerL1L2L3	Abbr.:		Mandatory	☐
FB Name:	Heating Rod			Can be internal	☐
Description					
The present power value for consumption (positive value) of the device cumulated at all phases L1, L2, L3.					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Power	DPT_ID:	14.056		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present power value for consumption at all phases	M	full	W	none
Access Type					
◆ Output					
	this → M	☒	this → 1	☐	
	Spontaneous	☒	COV:	☒	Δ-Value: >1 %
			Cyclic	☐	Period:
	Request	☒			
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
	Default Group Address:	none			
◆ Property					Mandatory: ☒
	DP Address: (in the server)	IO Type(ID):	141	Property ID:	64
		Start-Index:	1	N° of elements	1
	Property access:	Read only	☒	Read/Write	☐
	Protection *)	Read level	-	Write level	-
Dynamics					
	Power down:	Save:	☐		
	Power up:	Value:	No initialisation:	☒	Default value:
			Saved value:	☐	Current value (not for input):
		Transmit on bus (only for output):		☐	Read from bus (only for input):
Exception Handling					
Special Features					

1.4.4.3 Parameter: Heating Rod

1.4.4.3.1 LoadMin

FB:		Property Name (Server):	LoadMin			Mandatory	☐	Optional	☒
Description:									
Low boundary of the load									
DPT:	Name	DPT Value Power	DPT_ID	14.056	Datatype format:				
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Float	Parameter setting the low boundary of the load		M	full	W	--	none		
Communication:									
DP Address: (in the server)		Object Type:	141	Property ID:		111			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☐	-	Read/Write ☒					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☐ Curr Value ☒ Default Value ☐									
Special Features:									
None.									

1.4.4.4 Diagnostic Data: Heating Rod

1.4.4.4.1 PowerL1Max

FB:		Property Name (Server):	PowerL1Max			Mandatory	☐	Optional	☒
Description:									
The present maximum power value for consumption (positive value) of the device at phase L1.									
DPT:	Name	DPT Value Power	DPT_ID	14.056	Datatype format:				
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Float	Value informing about the maximum power value for consumption at L1		M	full	W	--	none		
Communication:									
DP Address: (in the server)		Object Type:	141	Property ID:		120			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.4.4.4.2 PowerL2Max

FB:		Property Name (Server):	PowerL2Max			Mandatory	☐	Optional	☒
Description:									
The present maximum power value for consumption (positive value) of the device at phase L2.									
DPT:	Name	DPT_Value_Power	DPT_ID	14.056	Datatype format:	14.056			
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Float	Value informing about the maximum power value for consumption at L2		M	full	W	--	none		
Communication:									
DP Address: (in the server)		Object Type:	141	Property ID:		121			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.4.4.4.3 PowerL3Max

FB:		Property Name (Server):	PowerL3Max			Mandatory	☐	Optional	☒
Description:									
The present maximum power value for consumption (positive value) of the device at phase L3.									
DPT:	Name	DPT_Value_Power	DPT_ID	14.056	Datatype format:	14.056			
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Float	Value informing about the maximum power value for consumption at L1		M	full	W	--	none		
Communication:									
DP Address: (in the server)		Object Type:	141	Property ID:		122			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.4.4.4.4 PowerL1L2L3Max

FB:		Property Name (Server):	PowerL1L2L3Max			Mandatory	☐	Optional	☒
Description:									
The present maximum power value for consumption (positive value) of the device cumulated at all phases L1, L2, L3.									
DPT:	Name	DPT Value	Power	DPT_ID		Datatype format:			
Field	Description			Sup.	Range	Unit	Resol.:	Default	
Float	Value informing about the maximum power value for consumption at all phases			M	full	W	--	none	
Communication:									
DP Address: (in the server)		Object Type:		141	Property ID:		123		
		Start-Index:		1	N° of elements		1		
Property access:		Read only ☒		-	Read/Write ☐				
Protection		Read level		-	Write level		-		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.4.4.4.5 EnergyEfficiency

FB:		Property Name (Server):	EnergyEfficiency			Mandatory	☐	Optional	☒
Description:									
Energy efficiency i.e. heating load output / electrical load input									
DPT:	Name	DPT Scaling		DPT_ID	5.001	Datatype format:			
Field	Description			Sup.	Range	Unit	Resol.:	Default	
Unsigned Value	Value informing about the energy efficiency			M	full	%	0,04	none	
Communication:									
DP Address: (in the server)		Object Type:		141	Property ID:		124		
		Start-Index:		1	N° of elements		1		
Property access:		Read only ☒		-	Read/Write ☐				
Protection		Read level		-	Write level		-		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.5 Functional Block: Buffer Storage

A buffer storage is used to store surplus energy from heat generator such as heat pumps.

With a buffer storage, a surplus of thermal energy can be stored as thermal energy. Commonly water is used to buffer the thermal energy.

This allows that the heat energy can be consumed at a later stage when it is demanded.

For energy management reasons the temperature of this buffer can be increased or decreased

1.5.1 Function Block Diagram: Buffer Storage

Buffer storage			
Inputs	Parameter	Description	Output
TempWaterSetp	TempWaterSetp		
		TempWaterSetpEff	TempWaterSetpEff
		BufferFull	BufferFull
		TempWaterTop	TempWaterTop
		TempWater2nd	TempWater2nd
		TempWater3rd	TempWater3rd
		TempWater4th	TempWater4th
		TempWater5th	TempWater5th
	BufferFullHyst		
	TempHeatMax		
	TempHeatMin		
	TempCoolMax		
	TempCoolMin		
		TempLoss	
		BufferVolume	
		BufferType	

Figure 5 – Buffer Storage functional block

1.5.2 List of Datapoints: Buffer Storage

Inputs		
Datapoint	Description/Remarks	Datapoint Type
TempWaterSetp	Temperature Setpoint of the buffer storage	9.001 DPT_Value_Temp

Outputs		
Datapoint	Description/Remarks	Datapoint Type
TempWaterSetpEff	Feedback of the present temperature setpoint of the buffer storage set by the input data point or by internal settings	9.001 DPT_Value_Temp
BufferFull	This signal is sent from the buffer, where the heating rod is located. As long the buffer is full (1) the heating rod is switched off to avoid over temperature in the buffer	1.002 DPT_Bool
TempWaterTop	Present temperature of the buffer storage top layer	9.001 DPT_Value_Temp
TempWater2nd	Present temperature of the buffer storage second layer, if not available the Value of the top level could be used	9.001 DPT_Value_Temp
TempWater3rd	Present temperature of the buffer storage third layer, if not available the Value of the second level could be used	9.001 DPT_Value_Temp
TempWater4th	Present temperature of the buffer storage fourth layer, if not available the Value of the third level could be used	9.001 DPT_Value_Temp
TempWater5th	Present temperature of the buffer storage fifth layer, if not available the Value of the fourth level could be used	9.001 DPT_Value_Temp

Parameters		
Datapoint	Description/Remarks	Datapoint Type
BufferFullHyst	Hysteresis of the BufferFull signal	9.002 DPT_Value_Tempd
TempHeatMax	maximum buffer storage temperature for heating	9.001 DPT_Value_Temp
TempHeatMin	minimum buffer storage temperature for heating	9.001 DPT_Value_Temp
TempCoolMax	maximum buffer storage temperature for cooling	9.001 DPT_Value_Temp
TempCoolMin	minimum buffer storage temperature for cooling	9.001 DPT_Value_Temp

Diagnostic Data		
Datapoint	Description/Remarks	Datapoint Type
TempLoss	Temperature loss per hour in percent given by the datasheet of buffer storage. As the ratio is fixed to 1 hour a percentage value is used as DPT	5.001 DPT_Scaling
BufferVolume	Size of the buffer in liters given by the datasheet of buffer storage	12.1200 DPT_VolumeLiquid_Litre
BufferType	Type of Buffer 1 = Buffer storage is used for heating, 0 = Buffer storage is used for cooling	1.100 DPT_Heat/Cool

1.5.3 Inputs, Outputs, Parameters and Diagnostic Data of Buffer Storage

1.5.3.1 Inputs: Buffer Storage

1.5.3.1.1 TempWaterSetp

DP Name:	TempWaterSetp	Abbr.:		Mandatory	☒
FB Name:	Buffer Storage			Can be internal	☐
Description					
Temperature Setpoint of the top layer buffer storage					
Datapoint Type					
DPT Name:					
DPT Format:	DPT Value Temp	DPT ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Setting the temperature setpoint of the top layer of the buffer storage	M	5 to 95	°C	65°C
Access Type					
◆ Input					
N → this	☒	1 → this	☐		
Spontaneous	☒	Cyclically:	☐	Time-out:	
Request	☐	Polling:	☐	Period:	
Communication Type					
◆ Group Object Datapoint				Mandatory:	☒
Default Group Address:		None			
◆ Property				Mandatory:	☒
DP Address: (in the server)	IO Type(ID):	142	Property ID:	51	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☐	Read/Write	☒	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Value below or above the limit shall be ignored.					
Special Features					

1.5.3.2 Outputs: Buffer Storage

1.5.3.2.1 TempWaterSetpEff

DP Name:	TempWaterSetpEff	Abbr.:		Mandatory	☒
FB Name:	Buffer Storage			Can be internal	☐
Description					
Feedback of the present temperature setpoint of the buffer storage set by the input data point or by internal settings					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT_ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present temperature setpoint of the buffer storage	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	142	Property ID:	60	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.5.3.2.2 BufferFull

DP Name:	BufferFull	Abbr.:		Mandatory	☒
FB Name:	Buffer Storage			Can be internal	☐
Description					
This datapoint indicates when TRUE that the buffer temperature has reached the value of temperature setpoint of the water. To have a stable output this value remains TRUE as long the buffer temperature is higher than temperature setpoint of the water minus hysteresis of the buffer. If the buffer temperature is below temperature setpoint of the water minus hysteresis of the buffer the value is FALSE and remains add FALSE until the buffer temperature has reached the value of temperature setpoint of the water					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Bool	DPT ID:	1.002		
Field		Supp.	Range	Unit	Default
b	Value informing if the buffer temperature has reached the value of the temperature setpoint of the water (1) or not (0)	M	full	bool	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	Min repetition time:
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:		none			
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	142	Property ID:	61	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.5.3.2.3 TempWaterTop

DP Name:	TempWaterTop	Abbr.:		Mandatory	☒
FB Name:	Buffer Storage			Can be internal	☐
Description					
Present temperature of the buffer storage top layer					
Datapoint Type					
DPT Name:					
DPT Format:	DPT_Value_Temp	DPT ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present temperature of the buffer storage top layer	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	142	Property ID:	62	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling					
Special Features					

1.5.3.2.4 TempWater2nd

DP Name:	TempWater2nd		Abbr.:		Mandatory	☒
FB Name:	Buffer Storage				Can be internal	☐
Description						
Present temperature of the buffer storage second layer, if not available the value of the top level could be used						
Datapoint Type						
DPT_Name:						
DPT Format:	DPT_Value_Temp		DPT_ID:	9.001		
Field		Supp.	Range	Unit	Default	
Float	Value informing about the present temperature of the buffer storage second layer		M	full	°C	none
Access Type						
◆ Output						
	this → M	☒	this → 1	☐		
	Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
			Cyclic	☐	Period:	
	Request	☒				
Communication Type						
◆ Group Object Datapoint					Mandatory:	☒
	Default Group Address:	none				
◆ Property					Mandatory:	☒
	DP Address: (in the server)	IO Type(ID):	142	Property ID:	63	
		Start-Index:	1	N° of elements	1	
	Property access:	Read only	☒	Read/Write	☐	
	Protection *)	Read level	-	Write level	-	
Dynamics						
	Power down:	Save:	☐			
	Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐	
		Transmit on bus (only for output):	☐	Read from bus (only for input):	☐	
Exception Handling						
Special Features						

1.5.3.2.5 TempWater3rd

DP Name:	TempWater3rd	Abbr.:		Mandatory	☒
FB Name:	Buffer Storage			Can be internal	☐
Description					
Present temperature of the buffer storage 3rd layer, if not available the value of the second level could be used					
Datapoint Type					
DPT_Name:	DPT_Value_Temp				
DPT Format:		DPT_ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present temperature of the buffer storage third layer	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	142	Property ID:	64	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.5.3.2.6 TempWater4th

DP Name:	TempWater4th	Abbr.:		Mandatory	☒
FB Name:	Buffer Storage			Can be internal	☐
Description					
Present temperature of the buffer storage 4th layer, if not available the value of the 3rd level could be used					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Temp	DPT_ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present temperature of the buffer storage fourth layer	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	142	Property ID:	65	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.5.3.2.7 TempWater5th

DP Name:	TempWater5th	Abbr.:		Mandatory	☒
FB Name:	Buffer Storage			Can be internal	☐
Description					
Present temperature of the buffer storage 5th layer, if not available the value of the 4th level could be used					
Datapoint Type					
DPT_Name:					
DPT Format:	DPT_Value_Temp	DPT_ID:	9.001		
Field		Supp.	Range	Unit	Default
Float	Value informing about the present temperature of the buffer storage fifth layer	M	full	°C	none
Access Type					
◆ Output					
this → M	☒	this → 1	☐		
Spontaneous	☒	COV:	☒	Δ-Value:	>1 %
		Cyclic	☐	Period:	
Request	☒				
Communication Type					
◆ Group Object Datapoint					Mandatory: ☒
Default Group Address:	none				
◆ Property					Mandatory: ☒
DP Address: (in the server)	IO Type(ID):	142	Property ID:	66	
	Start-Index:	1	N° of elements	1	
Property access:	Read only	☒	Read/Write	☐	
Protection *)	Read level	-	Write level	-	
Dynamics					
Power down:	Save:	☐			
Power up:	Value:	No initialisation:	☒	Default value:	☐
		Saved value:	☐	Current value (not for input):	☐
	Transmit on bus (only for output):		☐	Read from bus (only for input):	☐
Exception Handling					
Special Features					

1.5.3.3 Parameter: Buffer Storage

1.5.3.3.1 BufferFullHyst

FB:		Property Name (Server):	BufferFullHyst				Mandatory	☒
							Optional	☐
Description:								
Hysteresis of the "BufferFull" signal								
DPT:	Name	DPT_Value_Tempd	DPT_ID		Datatype format:	9.002		
Field	Description		Sup.	Range	Unit	Resol.:	Default	
Float	Parameter setting the hysteresis of the buffer full signal		M	5 to 25	K	0,01	10	
Communication:								
DP Address: (in the server)		Object Type:	142	Property ID:		111		
		Start-Index:	1	N° of elements		1		
Property access:		Read only ☒	-	Read/Write ☐				
Protection		Read level	-	Write level		-		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐								
Value below or above the limit must be set to the limit								
Special Features:								
None.								

1.5.3.3.2 TempHeatMax

FB:		Property Name (Server):	TempHeatMax				Mandatory	☒
							Optional	☐
Description:								
maximum buffer storage temperature for heating								
DPT:	Name	DPT_Value_Temp	DPT_ID		Datatype format:	9.001		
Field	Description		Sup.	Range	Unit	Resol.:	Default	
Float	Parameter setting the maximum buffer storage temperature for heating		M	full	°C	0,01	none	
Communication:								
DP Address: (in the server)		Object Type:	142	Property ID:		112		
		Start-Index:	1	N° of elements		1		
Property access:		Read only ☒	-	Read/Write ☐				
Protection		Read level	-	Write level		-		
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐								
Special Features:								
None.								

1.5.3.3.3 TempHeatMin

FB:		Property Name (Server):	TempHeatMin			Mandatory	☐	Optional	☒
Description:									
Minimum buffer storage temperature for heating									
DPT:	Name	DPT_Value_Temp	DPT_ID		Datatype format:	9.001			
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Float	Parameter setting the minimum buffer storage temperature for heating		M	full	°C	0,01	none		
Communication:									
DP Address: (in the server)		Object Type:	142	Property ID:		113			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.5.3.3.4 TempCoolMin

FB:		Property Name (Server):	Temperature cooling maximum			Mandatory	☐	Optional	☒
Description:									
Minimum buffer storage temperature for cooling									
DPT:	Name	DPT_Value_Temp	DPT_ID		Datatype format:	9.001			
Field	Description		Sup.	Range	Unit	Resol.:	Default		
Float	Parameter setting the minimum storage temperature for cooling		M	full	°C	0,01	none		
Communication:									
DP Address: (in the server)		Object Type:	142	Property ID:		115			
		Start-Index:	1	N° of elements		1			
Property access:		Read only ☒	-	Read/Write ☐					
Protection		Read level	-	Write level		-			
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐									
Special Features:									
None.									

1.5.3.3.5 TempCoolMax

FB:		Property Name (Server):	TempCoolMax				Mandatory	☐	Optional	☒
Description:										
maximum buffer storage temperature for cooling										
DPT:	Name	DPT Value Temp	DPT_ID		Datatype format:	9.001				
Field	Description		Sup.	Range	Unit	Resol.:	Default			
Float	Parameter setting the maximum temperature for cooling		M	full	°C	0,01	none			
Communication:										
DP Address: (in the server)		Object Type:	142	Property ID:		114				
		Start-Index:	1	N° of elements		1				
Property access:		Read only ☒	-	Read/Write ☐						
Protection		Read level	-	Write level		-				
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐										
Special Features:										
None.										

1.5.3.4 Diagnostic Data: Buffer Storage

1.5.3.4.1 TempLoss

FB:		Property Name (Server):	TempLoss				Mandatory	☐	Optional	☒
Description:										
Temperature loss per hour in percent given by the datasheet of the buffer storage										
DPT:	Name	DPT Scaling	DPT_ID		Datatype format:	5.001				
Field	Description		Sup.	Range	Unit	Resol.:	Default			
Unsigned Value	Value informing about temperature loss per hour in percent		M	full	%	0,04	none			
Communication:										
DP Address: (in the server)		Object Type:	142	Property ID:		120				
		Start-Index:	1	N° of elements		1				
Property access:		Read only ☒	-	Read/Write ☐						
Protection		Read level	-	Write level		-				
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐										
Special Features:										
None.										

1.5.3.4.2 BufferVolume

FB:		Property Name (Server):	BufferVolume	Mandatory	☐	Optional	☒
Description:							
Size of the buffer in Liters given by the datasheet of buffer storage							
DPT:	Name	DPT_VolumeLiquid_Litre	DPT_ID		Datatype format:	12.1200	
Field	Description		Sup.	Range	Unit	Resol.:	Default
Unsigned Value	Value informing about the size of the buffer in liters		M	full	l	1	none
Communication:							
DP Address: (in the server)		Object Type:	142	Property ID:		121	
		Start-Index:	1	N° of elements		1	
Property access:		Read only ☒	-	Read/Write ☐			
Protection		Read level	-	Write level		-	
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Special Features:							
None.							

1.5.3.4.3 BufferType

FB:		Property Name (Server):	BufferType	Mandatory	☐	Optional	☒
Description:							
Type of Buffer: 1 = Buffer storage is used for heating, 0 = Buffer storage is used for cooling							
DPT:	Name	DPT_Heat/Cool	DPT_ID		Datatype format:	1.100	
Field	Description		Sup.	Range	Unit	Resol.:	Default
b	Value informing if buffer is used for heating (1) or for cooling (0)		M	full	bool	none	none
Communication:							
DP Address: (in the server)		Object Type:	142	Property ID:		122	
		Start-Index:	1	N° of elements		1	
Property access:		Read only ☒	-	Read/Write ☐			
Protection		Read level	-	Write level		-	
Exception Handling: Value after Power-up: Stored Value ☒ Curr Value ☐ Default Value ☐							
Special Features:							
None.							

1.6 Functional Block: Domestic Hot Water

A specific type of buffer storage is the Domestic Hot Water (DHW) buffer (DHW Buffer storage). It serves to store hot water to fixtures like a sink, shower, tub and any other appliance where water may be consumed or used by humans.

The water is stored in a buffer. For Energy management reasons the temperature of this buffer can be increased or decreased by the means of the Domestic Hot Water temperature setpoint or by the Domestic Hot Water operation mode command.

These function blocks already exist in the KNX Standard see Application Description Domestic Hot Water Control 7/11/3.

The function consists of the following functional blocks

- Domestic Hot Water Setpoint Manager (DHWSM)
- Domestic Hot Water Controller (DHWC) and
- HW Temperature Sensor (DHWTS)

For better fitting to energy management some datapoints and properties will be added in DHWC. These are:

- DHWBufferFull
- DHWBufferFullHyst
- Volume of the DHW buffer
- TempSetpointDHWMax

This extension will be described in an application note.